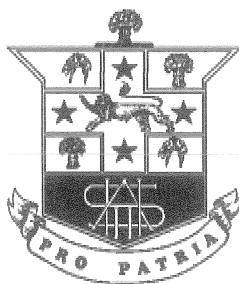


# HURLSTONE AGRICULTURAL HIGH SCHOOL



## 2014 MATHEMATICS HSC TRIAL EXAMINATION (ASSESSMENT TASK 4)

EXAMINERS ~ S. FAULDS, S. HACKETT, P. BICZO, G. HUXLEY, D. CRANCHER

### GENERAL INSTRUCTIONS

- Reading Time – 5 minutes.
- Working Time – 3 hours.
- Attempt **all** questions.
- Board approved calculators and mathematical templates may be used.
- This examination must **NOT** be removed from the examination room.
- Question 1 – 10 are to be completed on the Multiple Choice Answer Sheet.
- Show all necessary working in Questions 11 – 16.
- **Start each question in a separate answer booklet.**
- Put your student number on each booklet.
- A table of standard integrals is on the back of the Multiple Choice Answer Sheet.

**Total marks – 100**

### Section I

**10 marks**

- **Attempt Questions 1 – 10.**
- Allow about 15 minutes for this section.
- Fill in your answers on the multiple choice answer sheet provided.

### Section II

**90 marks**

- **Attempt Questions 11 – 16.**  
Each of these six (6) questions worth 15 marks. Allow about 2 hours and 45 minutes for this section. **Each question is to be started in a new answer booklet.** Additional booklets are available if required.

STUDENT NAME: \_\_\_\_\_

CLASS TEACHER: \_\_\_\_\_

## Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

---

1.  $|-6| - |-12| =$

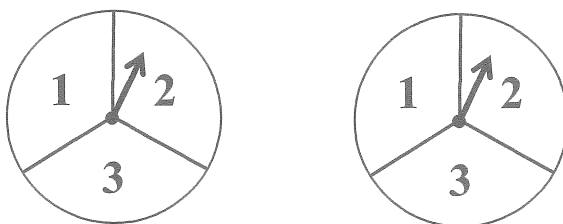
A: 6

B: -6

C: -18

D: 18

2.



Two identical spinners, containing the values 1, 2, and 3 are spun and the results on each are multiplied together. What is the probability that the resulting ~~sum~~<sup>product</sup> is either an even number or a number greater than 6?

A:  $\frac{1}{3}$

B:  $\frac{5}{9}$

C:  $\frac{4}{9}$

D:  $\frac{2}{3}$

3.  $\frac{\log_3 8}{\log_3 2} =$

A:  $2\log_3 2$

B:  $\log_3 6$

C: 4

D: 3

4. Fully factorised,  $16x^3 - 54 =$

A:  $2(2x - 3)(4x^2 + 12x + 9)$

B:  $2(2x - 3)(4x^2 + 6x + 9)$

C:  $2(2x - 3)(4x^2 - 6x + 9)$

D:  $2(2x - 3)(4x^2 - 12x + 9)$

5. If  $2\sqrt{80} + \sqrt{45} = a\sqrt{b}$ , then

A:  $a = 11, b = 5$

B:  $a = 5, b = 5$

C:  $a = 7, b = 5$

D:  $a = 17, b = 5$

6.

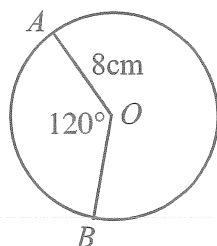


DIAGRAM  
IS NOT TO  
SCALE

Which of these calculations would NOT give the correct area of sector  $AOB$ ?

A:  $\frac{1}{2} \times 8^2 \times \frac{2\pi}{3}$

B:  $\frac{120}{360} \times \pi \times 8^2$

C:  $\frac{1}{2} \times 8^2 \times 120$

D:  $\frac{1}{2} \times 8^2 \times \frac{120}{180} \times \pi$

7. A primitive function of  $(3x-2)^3$  is:

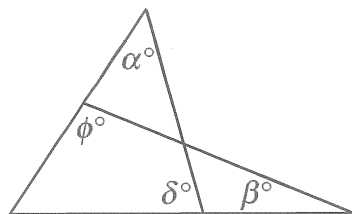
A:  $9(3x-2)^2$

B:  $\frac{(3x-2)^4}{12}$

C:  $\frac{(3x-2)^4}{4}$

D:  $9(3x-2)^4$

8. In the diagram below:



$\alpha^\circ + \delta^\circ =$

A:  $\beta^\circ$

B:  $180^\circ - (\beta^\circ + \phi^\circ)$

C:  $\phi^\circ$

D:  $\beta^\circ + \phi^\circ$

9. The exact value of  $\sin \frac{4\pi}{3} =$

A:  $\frac{1}{2}$

B:  $-\frac{1}{2}$

C:  $\frac{\sqrt{3}}{2}$

D:  $-\frac{\sqrt{3}}{2}$

10.  $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4 \sin 2x \, dx =$

A: 4

B: -4

C: 0

D: 2

## Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

*Answer each question in a new answer booklet.*

All necessary working should be shown in every question.

---

| Question 11 (15 marks) | Start a new answer booklet.                                                           | Marks |
|------------------------|---------------------------------------------------------------------------------------|-------|
| (a)                    | Solve: $ x - 5  \geq 2$                                                               | 2     |
| (b)                    | Find the exact value (in simplest form) of $\sqrt{p^4 - 2p^2}$ when $p = 2\sqrt{5}$ . | 2     |
| (c)                    | State the range of $f(x) = (x + 2)^2 - 3$ .                                           | 1     |
| (d)                    | Evaluate $\sqrt{\frac{30}{7} - \sqrt{12}}$ correct to 3 significant figures.          | 1     |
| (e)                    | Express $0.6\dot{4}$ as a fraction in simplest form, showing all working.             | 3     |
| (f)                    | Solve: $\frac{x}{8} - \frac{x-6}{4} = 2$                                              | 3     |
| (g)                    | Given $5^m = 4$ , evaluate $5^{1-2m}$ .                                               | 3     |

**Question 12 (15 marks) Start a new answer booklet.**

**Marks**

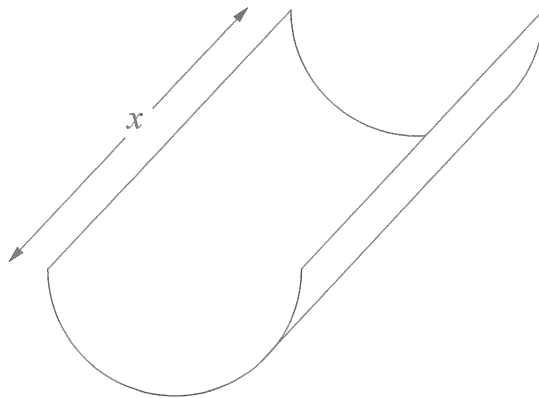
- (a) Use the definition of the derivative,  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ , to find  $f'(x)$  when  $f(x) = \frac{1}{x}$ . 2

- (b) (i) Use the quotient rule to show that if  $y = \frac{x}{(2x-3)^3}$ ,  $\frac{dy}{dx} = \frac{-4x-3}{(2x-3)^4}$ . 2

- (ii) Hence, find the equation of the normal to the curve  $y = \frac{x}{(2x-3)^3}$ , at the point  $(1, -1)$ . 2

- (c) Given  $f'(x) = x(x-3)(x+1)$  and  $f(0) = 0$ , find the function  $f(x)$ . 2

(d)



The diagram above shows a half-pipe, which is to be made from a rectangular piece of metal of length  $x$  m. The metal sheet is to be fabricated to form the open half cylinder shown. The perimeter of the rectangular sheet is 60 m.

- (i) By using Calculus, find the dimensions of the rectangle that will give the maximum surface area. 2
- (ii) For the dimensions giving this maximum area, find the height from the ground up to the top of the half-pipe. 1

**Question 12 continues on the next page...**

**Question 12** (continued)

**Marks**

- (e) A continuous curve,  $y = f(x)$  has the following properties for the interval  $a \leq x \leq b$ .

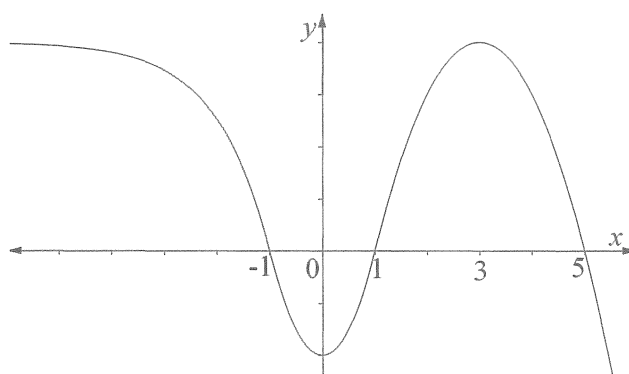
**2**

$$f(x) < 0, f'(x) < 0, f''(x) > 0$$

Sketch a curve satisfying these conditions for  $a \leq x \leq b$ .

- (f)

**2**



Sketch the derivative function of the curve above.

| Question 13 (15 marks)    Start a new answer booklet. |                                                                                                                                                                                                                              | Marks |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| (a)                                                   | Solve: $\sin^2 x - \sin x - 2 = 0$ for $-\pi \leq x \leq \pi$ .                                                                                                                                                              | 3     |
| (b)                                                   | Show that: $\frac{2\cos^3 \theta - \cos \theta}{\sin \theta \cos^2 \theta - \sin^3 \theta} = \cot \theta$ .                                                                                                                  | 3     |
| (c)                                                   | (i) Sketch the curve $y = 3\sin 2x$ in the domain $0 \leq x \leq 2\pi$ showing the main features of the graph.                                                                                                               | 2     |
|                                                       | (ii) Use your graph to find the number of solutions to the equation $3\sin 2x - 1 = 0$ for $0 \leq x \leq 2\pi$ .                                                                                                            | 1     |
| (d)                                                   | Find: $\frac{d}{dx} \left( \tan^2 \left( \frac{x}{3} \right) \right)$                                                                                                                                                        | 2     |
| (e)                                                   | Ship <b>A</b> sails from port <b>P</b> on a bearing of $060^\circ$ for 56 nautical miles. Ship <b>B</b> sails from port <b>P</b> on a bearing of $110^\circ$ for 48 nautical miles. Draw a diagram to show this information. |       |
|                                                       | (i) Calculate the distance between the ships, correct to one decimal place.                                                                                                                                                  | 2     |
|                                                       | (ii) Find the bearing of ship <b>A</b> from ship <b>B</b> .                                                                                                                                                                  | 2     |

Question 14 (15 marks) Start a new answer booklet.

Marks

- (a) Find the values of  $m$  for which the equation  $x^2 + (m - 2)x + 4 = 0$  has no real roots. 2
- (b) The quadratic equation  $2x^2 - (m + 2)x + m = 0$  has one root which is twice the other. Find the values of  $m$ . 2
- (c) Differentiate with respect to  $x$ :  $e^x \sin x$  1
- (d) Find the primitive of the function  $g(x) = \frac{x}{x^2 - 11}$ . 2

(e)

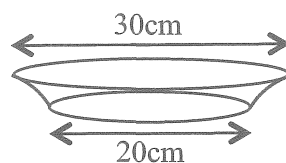
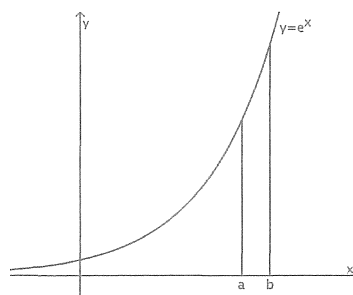


Diagram is not to scale

An artist produces a brass bowl as shown above. The artist made the following comment: "I received inspiration from the mathematical curve  $y = e^x$ . If the curve is rotated about the  $x$ -axis, the volume formed will be exactly that of the bowl."

- (i) If the curve  $y = e^x$  is rotated about the  $x$ -axis, the domain of the section of curve that is rotated to form the bowl is  $a \leq x \leq b$ . 2



Graph not to scale

Find values for  $a$  and  $b$  to provide a bowl with the given dimensions.

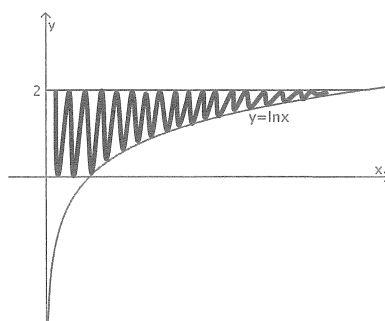
- (ii) What will be the *exact* capacity of the bowl in mL? 3

Question 14 continues on the next page...

Question 14 (continued)

Marks

- (f) Simpson's Rule is to be used to approximate the area enclosed by the  $y$ -axis, the curve  $y = \ln x$  and the lines  $y = 0$  and  $y = 2$  as shown below:



- (i) Copy and complete the table below in your answer booklet. Write your calculated values to 1 decimal place.

1

|     |   |     |     |     |   |
|-----|---|-----|-----|-----|---|
| $y$ | 0 | 0.5 | 1   | 1.5 | 2 |
| $x$ | 1 | 1.6 | 2.7 |     |   |

- (ii) Use two applications of Simpson's Rule to find an approximation for the area described. Give your answer correct to one decimal place.

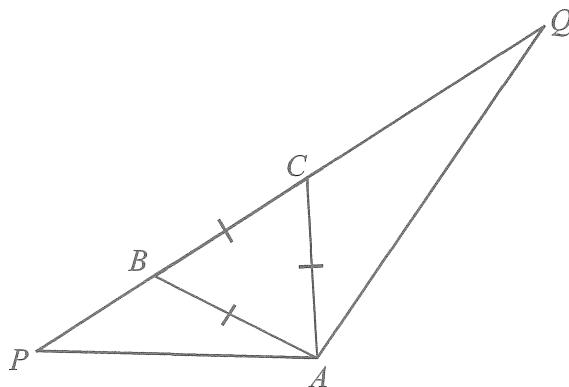
2

**Question 15 (15 marks)**      **Start a new answer booklet.**

**Marks**

- (a) (i) A point  $P(x, y)$  is equidistant from the points  $A(-7, 4)$  and  $B(-1, 12)$ . 2  
 Show that the locus of the point  $P$  is  $3x + 4y - 20 = 0$ .
- (ii) Show that the equation of the locus found in (i) is also the equation of a focal chord of the parabola  $(x - 4)^2 = 4(y - 1)$ . 3
- (iii) Show that the focal chord and the parabola intersect on the  $y$ -axis at a point  $G$ . 2
- (iv) Show that the equation of the line perpendicular to the focal chord which passes through the focus,  $S$ , is  $4x - 3y - 10 = 0$  and find the point,  $H$ , where it crosses the  $x$ -axis. 2
- (v) Hence, find the area of  $\triangle GSH$ . 2

(b)



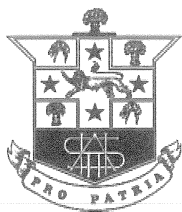
In the diagram,  $\triangle ABC$  is equilateral and  $\angle PAQ = 120^\circ$ .

- (i) Prove  $\triangle PBA \sim \triangle PAQ$ . 3
- (ii) Hence, show  $PA^2 = PQ \cdot PB$ , giving a reason for your answer. 1

| Question 16 (15 marks) | Start a new answer booklet.                                                                                                                                                                                                                                                                       | Marks |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| (a)                    | A banksia bush was planted when it was 50cm tall.<br>At the end of the first year after planting, the banksia was 90cm tall.<br>Each year's growth was then 80% of the previous year's growth.<br>What is the limiting maximum height of the banksia?                                             | 1     |
| (b)                    | The second term of an arithmetic series is 37 and the sixth term is 17.<br>What is the sum of the first ten terms of the series?                                                                                                                                                                  | 2     |
| (c)                    | Kylie invests \$ $P$ at 9% per annum compounded annually.<br>She plans to withdraw \$5000 at the end of each year for six years to cover university fees.                                                                                                                                         |       |
| (i)                    | Write down an expression for the amount \$ $A_1$ remaining in the account following the withdrawal of the first \$5000.                                                                                                                                                                           | 1     |
| (ii)                   | Find an expression for the amount \$ $A_3$ remaining in the account after the third withdrawal.                                                                                                                                                                                                   | 2     |
| (iii)                  | How much does Kylie need to invest if the account balance is to be \$0 at the end of the six years?                                                                                                                                                                                               | 2     |
| (d)                    | A game is played in which two coloured dice are thrown once.<br>The six faces of the red die are numbered 1, 3, 5, 7, 9 and 11.<br>The six faces of the white die are numbered 2, 4, 6, 8, 10 and 12.<br>The player wins if the number on the white die is larger than the number on the red die. |       |
| (i)                    | Show that the probability of the player winning a game is $\frac{7}{12}$ .                                                                                                                                                                                                                        | 1     |
| (ii)                   | What is the probability that the player wins exactly once in two successive games?                                                                                                                                                                                                                | 2     |
| (iii)                  | What is the probability that the player wins at least once in two successive games?                                                                                                                                                                                                               | 1     |
| (e)                    | There is a 75% chance that Lloyd will pass his driving test and a 40% chance that Harry will pass his.                                                                                                                                                                                            |       |
| (i)                    | Draw a probability tree to represent this information, including the list of possible outcomes.                                                                                                                                                                                                   | 2     |
| (ii)                   | Find the probability that only one of Lloyd and Harry pass the test.                                                                                                                                                                                                                              | 1     |

**END OF EXAMINATION**

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# HURLSTONE AGRICULTURAL HIGH SCHOOL

Year 12 Mathematics 2014

HSC Trial Examination

(Assessment Task 4)

NAME.....

CLASS.....

**EXAMPLE**     $2 + 4 =$

A    2    B    4    C    6    D    8    (A) (B) ● (D)

***ATTEMPT ALL QUESTIONS***

|    |                 |
|----|-----------------|
| 1  | (A) (B) (C) (D) |
| 2  | (A) (B) (C) (D) |
| 3  | (A) (B) (C) (D) |
| 4  | (A) (B) (C) (D) |
| 5  | (A) (B) (C) (D) |
| 6  | (A) (B) (C) (D) |
| 7  | (A) (B) (C) (D) |
| 8  | (A) (B) (C) (D) |
| 9  | (A) (B) (C) (D) |
| 10 | (A) (B) (C) (D) |

**This sheet should be removed from the question booklet  
and handed in with your answer booklets.**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

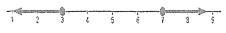
$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

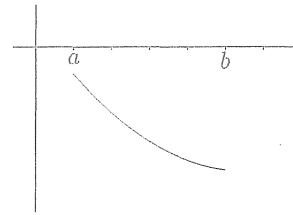
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

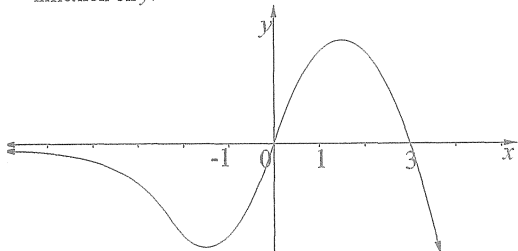
NOTE :  $\ln x = \log_e x, \quad x > 0$

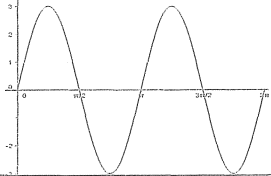
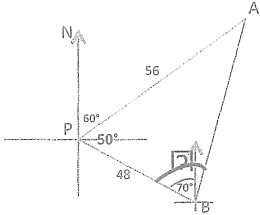
| Year 12 Mathematics Trial HSC (Task 4) 2014 |         |                         |
|---------------------------------------------|---------|-------------------------|
| Question No's. 1 – 10                       |         | Multiple Choice Answers |
|                                             | Answers |                         |
|                                             | 1. B    |                         |
|                                             | 2. D    |                         |
|                                             | 3. D    |                         |
|                                             | 4. B    |                         |
|                                             | 5. A    |                         |
|                                             | 6. C    |                         |
|                                             | 7. B    |                         |
|                                             | 8. D    |                         |
|                                             | 9. D    |                         |
|                                             | 10. C   |                         |

| QUESTION 11 –Advanced Mathematics Trial HSC Exam 2014 |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SAMPLE SOLUTION                                       |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| a)                                                    |  $x \leq 3, x \geq 7$                                                                                                                                                                                                              | <b>2 marks</b> – correct answer clearly showing how the answer was reached<br><b>1 mark</b> – substantial progress towards correct answer<br><i>This part was generally done well. A number of students found the critical values <math>x=3</math> and <math>x=7</math>, and then tested them in the original inequality - this step achieved nothing. The best responses utilized a number line to find all points more than 2 units away from 5.</i> |
| b)                                                    | $\begin{aligned}\sqrt{p^4 - 2p^2} &= \sqrt{(2\sqrt{5})^4 - 2(2\sqrt{5})^2} \\ &= \sqrt{400 - 2 \times 20} \\ &= \sqrt{360} \\ &= 6\sqrt{10}\end{aligned}$                                                                                                                                                             | <b>2 marks</b> – correct answer in simplest form<br><b>1 marks</b> – correct answer unsimplified<br><i>A range of errors were made in the simplification of the surds in this question. The question stated "in simplest form" yet many students failed to simplify their answer.</i>                                                                                                                                                                  |
| c)                                                    | All real $y, y \geq -3$                                                                                                                                                                                                                                                                                               | <b>1 mark</b> – correct answer<br><i>The best responses sketched the parabola, using the transformation rules, then read the range from their graph.</i>                                                                                                                                                                                                                                                                                               |
| d)                                                    | 0.906                                                                                                                                                                                                                                                                                                                 | <b>1 mark</b> – correct answer to 3 sig figs                                                                                                                                                                                                                                                                                                                                                                                                           |
| e)                                                    | <p>Let <math>x = 0.6\dot{4}</math><br/> <math>x = 0.6444...</math><br/> <math>10x = 6.444... \quad (1)</math><br/> <math>100x = 64.444... \quad (2)</math><br/> <math>(2) - (1) \quad 90x = 58</math><br/> <math>x = \frac{58}{90} = \frac{29}{45}</math><br/> <math>\therefore 0.6\dot{4} = \frac{29}{45}</math></p> | <b>3 marks</b> – correct answer in simplest form, showing all steps using a valid method<br><b>2 marks</b> – correct answer NOT in simplest form, showing all steps using a valid method<br><b>1 mark</b> – substantial progress towards correct answer<br><i>Many students failed to simplify the fraction or show all working, even though both of these were specified in the question.</i>                                                         |
| f)                                                    | $\begin{aligned}\frac{x}{8} - \frac{x-6}{4} &= 2 \quad (\times 8) \\ x - 2(x-6) &= 16 \\ x - 2x + 12 &= 16 \\ -x &= 4 \\ x &= -4\end{aligned}$                                                                                                                                                                        | <b>3 marks</b> – correct solution<br><b>2 marks</b> – substantial progress towards correct answer<br><b>1 mark</b> – limited progress towards correct answer<br><i>This question was done well by most students.</i>                                                                                                                                                                                                                                   |
| g)                                                    | $\begin{aligned}5^{1-2n} &= \frac{5^1}{5^{2n}} \\ &= \frac{5}{(5^n)^2} \\ &= \frac{5}{4^2} \\ &= \frac{5}{16}\end{aligned}$                                                                                                                                                                                           | <b>3 marks</b> – correct solution<br><b>2 marks</b> – substantial progress towards correct answer<br><b>1 mark</b> – limited progress towards correct answer<br><i>Many students complicated this question by using logarithms – often unsuccessfully.</i>                                                                                                                                                                                             |

| Year 12 Trial<br>Question No.12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mathematics<br>Solutions and Marking Guidelines                                                                                                                                                                                                                                                                                | Examination 2014                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Outcomes Addressed in this Question</b><br>P6 relates the derivative of a function to the slope of its graph<br>P7 determines the derivative of a function through routine application of the rules of differentiation<br>H5 applies appropriate techniques from the study of <b>calculus</b> , <b>geometry</b> , probability, trigonometry and series to solve problems<br>H6 uses the derivative to determine features of the graph of a function<br>H7 uses the features of a graph to deduce information about the derivative |                                                                                                                                                                                                                                                                                                                                |                                                                                          |
| Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Solutions                                                                                                                                                                                                                                                                                                                      | Marking Guidelines                                                                       |
| P7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $(a) f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \text{ where } f(x) = \frac{1}{x}$ $= \lim_{h \rightarrow 0} \frac{\frac{x - (x+h)}{x(x+h)}}{h}$ $= \lim_{h \rightarrow 0} \frac{-h}{x(x+h)} \times \frac{1}{h}$ $= \lim_{h \rightarrow 0} \frac{-1}{x(x+h)}$ $= \frac{-1}{x(x+0)} = \frac{-1}{x^2}$ | 2 marks : correct solution<br><br>1 mark : substantial progress towards correct solution |
| P7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $(b)(i) y = \frac{x}{(2x-3)^3}$ $\frac{dy}{dx} = \frac{(2x-3)^3 \times 1 - x \cdot 3(2x-3)^2 \cdot 2}{(2x-3)^6}$ $\frac{dy}{dx} = \frac{(2x-3)^2 ((2x-3) - 6x)}{(2x-3)^6}$ $\therefore \frac{dy}{dx} = \frac{-4x-3}{(2x-3)^4}$                                                                                                 | 2 marks : correct solution<br><br>1 mark : substantial progress towards correct solution |
| P6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $(ii) \text{ Gradient of tangent at } x=1 \text{ is } \frac{dy}{dx} = \frac{-4-3}{(2-3)^4} = -7$ $\therefore \text{ gradient of normal at } x=1 \text{ is } \frac{1}{7}$ $\therefore \text{ equation of normal is } y+1 = \frac{1}{7}(x-1)$ $\text{Equation is } x-7y-8=0.$                                                    | 2 marks : correct solution<br><br>1 mark : substantial progress towards correct solution |
| H5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $(c) f'(x) = x(x-3)(x+1)$ $= x(x^2 - 2x - 3)$ $= x^3 - 2x^2 - 3x$                                                                                                                                                                                                                                                              | 2 marks : correct solution<br><br>1 mark : substantial progress towards correct solution |

|    |                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                    |
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|    | $f(x) = \int (x^3 - 2x^2 - 3x) dx$ $f(x) = \frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2} + c$ $f(0) = 0, \therefore 0 = 0 - 0 - 0 + c$ $f(x) = \frac{x^4}{4} - \frac{2x^3}{3} - \frac{3x^2}{2}$                                                                                                                                                                                                    |                                                                                                    |
| H5 | $(d)(i) \begin{array}{l} x \quad \boxed{\phantom{000}} \quad y \\ 2x + 2y = 60 \\ x + y = 30 \\ y = 30 - x \\ A = x(30 - x) \end{array}$ $A = 30x - x^2$ $\frac{dA}{dx} = 30 - 2x = 0 \text{ for maximum area.}$ $\frac{d^2A}{dx^2} = -2$ $\text{Since } \frac{d^2A}{dx^2} < 0, \therefore \text{maximum when } x = 15$ $\therefore \text{maximum surface area when the rectangle is 15 m by 15 m.}$ | 2 marks : correct solution<br><br>1 mark : substantial progress towards correct solution           |
| H5 | $(ii) \begin{array}{l} \text{Height from the ground} \\ \text{equals the radius of the} \\ \text{semicircle.} \\ \text{Half the circumference} = 15 \\ \therefore \pi r = 15 \\ \therefore \text{radius} = \frac{15}{\pi} \\ \text{Height is } \frac{15}{\pi} \text{ or } 4.77 \text{ m} \end{array}$             | 1 mark: correct answer                                                                             |
| H6 | $(e) f(x) < 0: y \text{ values negative}$ $f'(x) < 0: \text{decreasing}$ $f''(x) > 0: \text{concave up}$                                                                                                                                                                                                        | 2 marks : correct solution<br><br>1 mark : displays graph with two of the given properties correct |

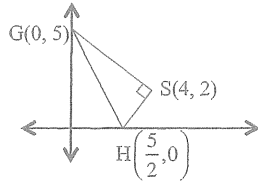
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                 |
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| H7 | <p>(f) At stationary points derivative = 0. <math>\therefore y' = 0</math> at <math>x = 0, 3</math>.<br/> Curve decreasing when <math>x &lt; 0</math> and <math>x &gt; 3</math>, and increasing when <math>0 &lt; x &lt; 3</math>.<br/> <math>\therefore y'</math> is negative when <math>x &lt; 0</math> and <math>x &gt; 3</math>, and <math>y'</math> positive when <math>0 &lt; x &lt; 3</math>.<br/> Maximum, minimum values of <math>y'</math> occur at points of inflexion on <math>y</math>.</p>  | <p>2 marks : correct solution</p> <p>1 mark : substantial progress towards correct solution</p> |
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| QUESTION 13 –Advanced Mathematics Trial HSC Exam 2014 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | SAMPLE SOLUTION                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| a)                                                    | $\sin^2 x - \sin x - 2 = 0$ $(\sin x - 2)(\sin x + 1) = 0$ $\sin x = 2 \quad \sin x = -1$ <p>no solution <math>x = -\frac{\pi}{2}</math></p> $\therefore x = -\frac{\pi}{2}$                                                                                                                                                                                                                                                                                        | <p>3 marks – correct solution</p> <p>2 marks – solution written in incorrect form (eg <math>-90^\circ</math>) or outside of the given domain (eg <math>\frac{3\pi}{2}</math>)</p> <p>1 mark – quadratic formed and correctly solved</p> <p>Many students made errors in solving the quadratic or obtaining the correct answer in the given domain. The best responses used the unit circle or a graph to obtain the correct boundary angle.</p> |
| b)                                                    | $\begin{aligned} LHS &= \frac{2\cos^2 \theta - \cos \theta}{\sin \theta \cos^2 \theta - \sin^2 \theta} \\ &= \frac{\cos \theta (2\cos^2 \theta - 1)}{\sin \theta (\cos^2 \theta - \sin^2 \theta)} \\ &= \frac{\cos \theta (2\cos^2 \theta - 1)}{\sin \theta (\cos^2 \theta - (1 - \cos^2 \theta))} \\ &= \frac{\cos \theta (2\cos^2 \theta - 1)}{\sin \theta (2\cos^2 \theta - 1)} \\ &= \frac{\cos \theta}{\sin \theta} \\ &= \cot \theta \\ &= RHS \end{aligned}$ | <p>3 marks – correct solution clearly showing all steps</p> <p>2 marks – substantial progress towards correct solution</p> <p>1 mark – original expression correctly factorised</p> <p>Some students failed to show all steps in their proof. All steps are essential in any "show that" response.</p>                                                                                                                                          |
| c) (i)                                                |                                                                                                                                                                                                                                                                                                                                                                                  | <p>2 marks – correct graph</p> <p>1 mark – graph with correct shape and correct period OR correct amplitude</p>                                                                                                                                                                                                                                                                                                                                 |
| c) (ii) 4 solutions                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>1 mark – correct answer</p> <p>The best responses drew the line <math>y=1</math> on their graph from part (i) to see how many points of intersection there were.</p>                                                                                                                                                                                                                                                                         |
| d)                                                    | $\begin{aligned} \frac{d}{dx} \left( \tan^2 \left( \frac{x}{3} \right) \right) &= 2 \tan \left( \frac{x}{3} \right) \sec^2 \left( \frac{x}{3} \right) \frac{1}{3} \\ &= \frac{2}{3} \tan \left( \frac{x}{3} \right) \sec^2 \left( \frac{x}{3} \right) \end{aligned}$                                                                                                                                                                                                | <p>2 marks – correct solution</p> <p>1 marks – substantial progress towards correct solution</p> <p>MANY students did not use the function of a function rule correctly, omitting one or more parts of the solution.</p>                                                                                                                                                                                                                        |
| e)                                                    |  <p>i)</p> $\begin{aligned} AB^2 &= 56^2 + 48^2 - 2 \times 56 \times 48 \times \cos 50^\circ \\ &= 1984.37 \\ AB &= 44.5 \text{ nm} \end{aligned}$ <p>ii)</p> $\frac{\sin \theta}{56} = \frac{\sin 50^\circ}{44.5}$ $\sin \theta = \frac{56 \sin 50^\circ}{44.5} = 0.964$ $\theta = 74^\circ 35'$ <p>Bearing = <math>74^\circ 35' - 70^\circ = 004^\circ 35'</math></p>        | <p>2 marks –correct solution</p> <p>1 marks – substantial progress towards correct solution</p> <p>Many students drew incorrect diagrams.</p> <p>Some misquoted the cosine rule.</p> <p>Students did not check their answers to ensure they made sense in the context of the question.</p>                                                                                                                                                      |

| Year 12 2014                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mathematics        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| Question No. 14                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Solutions and 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| Outcomes Addressed in this Question                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                    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| PE3 - solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations<br>H4 - expresses practical problems in mathematical terms based on simple given models<br>HE4 - uses the relationship between functions, inverse functions and their derivatives.<br>PE5 - determines derivatives which require the application of more than one rule of differentiation |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                    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| Outcome                                                                                                                                                                                                                                                                                                                                                                                                                            | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marking Guidelines 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|
| PE3                                                                                                                                                                                                                                                                                                                                                                                                                                | Question 14<br>a)<br>$x^2 + (m - 2)x + 4 = 0$<br>for no real roots $\Delta < 0$<br>$(m - 2)^2 - 4(1)(4) < 0$<br>$(m - 2)^2 - 4(1)(4) < 0$<br>$(m - 2)^2 - 4(1)(4) < 0$<br>$m^2 - 4m + 4 - 16 < 0$<br>$m^2 - 4m - 12 < 0$<br>$(m - 6)(m + 2) < 0$<br>$-2 < m < 6$                                                                                                                                                                                                                                                                                                             | 1 mark for stating<br>“for no real roots $\Delta < 0$ ”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                    | PE3<br>b)<br>$2x^2 - (m + 2)x + m = 0$<br>let the roots be $\alpha$ and $\beta$ such that $\beta = 2\alpha$<br>$\alpha + \beta = \frac{m + 2}{2}$<br>$\therefore \alpha + 2\alpha = \frac{m + 2}{2}, \text{ since } \beta = 2\alpha$<br>$\therefore 3\alpha = \frac{m + 2}{2}$<br>$\therefore \alpha = \left(\frac{m + 2}{6}\right) \dots \dots \dots (A)$<br>$\alpha\beta = \frac{m}{2}$<br>$\therefore \alpha(2\alpha) = \frac{m}{2}, \text{ since } \beta = 2\alpha$<br>$\therefore 2\alpha^2 = \frac{m}{2}$<br>$\therefore \alpha^2 = \frac{m}{4} \dots \dots \dots (B)$ | 1 mark for complete correct solution<br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br>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|            |                                                                                                                                                                                                                                                                |  |                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------|
|            | Continued from previous page...                                                                                                                                                                                                                                |  |                                                                                                          |
|            | Solve simultaneously A and B<br>Sub A into B to find $m$ $\left(\frac{m+2}{6}\right)^2 = \frac{m}{4}$ $\frac{m^2 + 4m + 4}{36} = \frac{m}{4}$ $m^2 + 4m + 4 = \frac{36m}{4}$ $m^2 + 4m + 4 = 9m$ $m^2 - 5m + 4 = 0$ $(m-1)(m-4) = 0$ $\therefore m = 1, m = 4$ |  | 1 mark for $m = 1, m = 4$                                                                                |
| PE5<br>HE4 | c) $\frac{d(e^x \cdot \sin x)}{dx} = e^x \cdot \cos x + \sin x \cdot e^x$ or<br>$e^x (\cos x + \sin x)$                                                                                                                                                        |  | 1 mark for correct differentiation                                                                       |
| HE4        | d)           The primitive of the function $g(x)$ is: $\int g(x) dx$ $= \int \frac{x}{x^2 - 11} dx$ $= \frac{1}{2} \int \frac{2x}{x^2 - 11} dx$ $= \frac{1}{2} \ln(x^2 - 11) + C$                                                                              |  | 2 marks for complete correct solution<br><br>1 mark for partial correct solution                         |
| H4         | e)           (i)           When $x = a, y = 10$ $\therefore 10 = e^a$ $\therefore a = \ln 10$ When $x = b, y = 15$ $\therefore 15 = e^b$ $\therefore b = \ln 15$                                                                                               |  | 2 marks for obtaining both $a$ and $b$ correctly<br><br>1 mark for obtaining either $a$ or $b$ correctly |
|            | The values are: $a = \ln 10, b = \ln 15$ .                                                                                                                                                                                                                     |  |                                                                                                          |

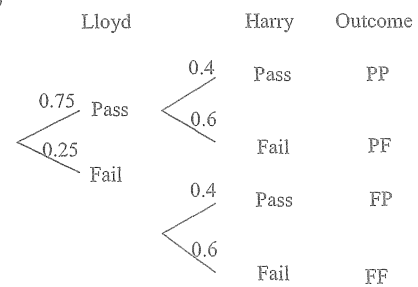
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                              |     |     |     |     |   |   |   |     |     |     |     |                                            |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|---|---|---|-----|-----|-----|-----|--------------------------------------------|
| <div>H4<br/>HE4<br/>PE5</div> | <div>(ii)</div> <div><math display="block">V = \pi \int_a^b y^2 dx</math><math display="block">= \pi \int_{\ln 10}^{\ln 15} (e^x)^2 dx</math><math display="block">= \pi \int_{\ln 10}^{\ln 15} e^{2x} dx</math><math display="block">= \pi \left[ \frac{e^{2x}}{2} \right]_{\ln 10}^{\ln 15}</math><math display="block">= \pi \left( \frac{e^{2 \cdot \ln 15}}{2} - \frac{e^{2 \cdot \ln 10}}{2} \right)</math><math display="block">= \pi \left( \frac{e^{\ln(15)^2}}{2} - \frac{e^{\ln(10)^2}}{2} \right)</math><math display="block">= \pi \left( \frac{225}{2} - \frac{100}{2} \right)</math><math display="block">= \frac{125\pi}{2} \text{ cm}^3</math><math display="block">= \frac{125\pi}{2} \text{ ml}</math></div> | <div>3 marks for complete correct solution</div> <div>2 marks for substantial progress to solution</div> <div>1 marks for limited progress to solution</div> |     |     |     |     |   |   |   |     |     |     |     |                                            |
| <div>H4</div>                 | <div>f)<br/>(i)</div> <div><table><tr><td>y</td><td>0</td><td>0.5</td><td>1</td><td>1.5</td><td>2</td></tr><tr><td>x</td><td>1</td><td>1.6</td><td>2.7</td><td>4.5</td><td>7.4</td></tr></table></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | y                                                                                                                                                            | 0   | 0.5 | 1   | 1.5 | 2 | x | 1 | 1.6 | 2.7 | 4.5 | 7.4 | <div>1 mark for both correct answers</div> |
| y                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.5                                                                                                                                                          | 1   | 1.5 | 2   |     |   |   |   |     |     |     |     |                                            |
| x                             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.6                                                                                                                                                          | 2.7 | 4.5 | 7.4 |     |   |   |   |     |     |     |     |                                            |
| <div>H4<br/>HE4</div>         | <div>(ii)</div> <div>Two applications of Simpson's rule:</div> <div><math display="block">A \approx \frac{\left(\frac{1}{2}\right)}{3} [f(0) + 4f(0.5) + f(1)] + \frac{\left(\frac{1}{2}\right)}{3} [f(1) + 4f(1.5) + f(2)]</math><math display="block">\approx \frac{1}{6} [1 + 4(1.6) + 2.7] + \frac{1}{6} [1 + 4(4.5) + 7.4]</math><math display="block">\approx 6.3</math><math display="block">\approx 6.4 \text{ units}^2 \text{ (to one decimal place)}</math></div>                                                                                                                                                                                                                                                     | <div>2 marks for complete correct solution [must apply Simpson's rule twice]</div> <div>1 mark for partial correct solution</div>                            |     |     |     |     |   |   |   |     |     |     |     |                                            |

| Year 12 Mathematics Trial HSC (Task 4) 2014      |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                 |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question No. 15 Solutions and Marking Guidelines |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                 |
| Outcomes Addressed in this Question              |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                 |
| H5                                               | applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                 |
| H2                                               | constructs arguments to prove and justify results                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                 |
| Outcome                                          | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                          | Marking Guidelines                                                                                                                                                              |
| H5                                               | (a) (i) $PA = PB$ $(x+7)^2 + (y-4)^2 = (x+1)^2 + (y-12)^2$ $x^2 + 14x + 49 + y^2 - 8y + 16 = x^2 + 2x + 1 + y^2 - 24y + 144$ $12x + 16y - 80 = 0$ Eqn. of locus: $3x + 4y - 20 = 0$ as required.                                                                                                                                                                                                                                   | 2 marks<br>Correct solution<br>1 mark<br>Substantial progress towards correct solution.                                                                                         |
| H5                                               | (ii) Parabola $(x-4)^2 = 4(y-1)$ Vertex (4, 1)<br>Concave up<br>Focal length = 1<br>$\therefore$ Focus (4, 2)<br><br>For focal chord, line must pass through focus<br>Sub. (4, 2) into equation of line from (i)<br>$3 \cdot 4 + 4 \cdot 2 - 20 = 12 + 8 - 20 = 0$<br>$\therefore$ Line passes through (4, 2), hence a focal chord.                                                                                                | 3 marks<br>Correct solution<br>2 mark<br>Substantial progress towards correct solution including description of parabola.<br>1 mark<br>Some progress towards a correct solution |
| H5                                               | (iii) To find y-intercepts, let $x = 0$<br>y-intercept of line $4y - 20 = 0$<br>$y = 5$<br>y-intercept of parabola $(-4)^2 = 4(y-1)$<br>$16 = 4y - 4$<br>$y = 5$<br>$\therefore$ Line and parabola meet on the y-axis at G when $y = 5$ .                                                                                                                                                                                          | 2 marks<br>Correct solution.<br>1 mark<br>Substantial progress towards correct solution.                                                                                        |
| H5                                               | (iv) Gradient of focal chord $3x + 4y - 20 = 0$ : $m = -\frac{3}{4}$<br><br>Gradient of perpendicular: $m = \frac{4}{3}$<br>Perpendicular passes through the point (4, 2) (ie. the focus)<br>$\therefore$ Equation of normal:<br>$y - 2 = \frac{4}{3}(x - 4)$<br>$3y - 6 = 4x - 16$<br>$4x - 3y - 10 = 0$<br>For x-intercept let $y = 0$<br>$4x = 10$<br>$x = \frac{5}{2}$<br><br>ie. H is the point $\left(\frac{5}{2}, 0\right)$ | 2 marks<br>Correct solution.<br>1 mark<br>Substantial progress towards correct solution including equation of normal.                                                           |
| H5                                               | (v)                                                                                                                                                                                                                                                                                                                                           | 2 marks<br>Correct solution.<br>1 mark<br>Substantial progress towards finding correct area.                                                                                    |

|               |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                       |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Using Pythagoras' Theorem:                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                       |
|               | $GS = 5 \text{ units}$<br>$HS = \frac{5}{2} \text{ units}$<br>$\therefore \text{Area } \triangle GSH = \frac{1}{2} \times \frac{5}{2} \times 5$<br>$= \frac{25}{4} \text{ units}$                                                                                                                                                                                                          |                                                                                                                                                                                                                       |
| <b>H2, H5</b> | <b>(b) (i)</b><br>$\angle CBA = 60^\circ$ (Angle in equilateral $\triangle ABC$ )<br>$\angle PBA = 180^\circ - \angle CBA$ (angles on a straight line)<br>$= 180^\circ - 60^\circ$<br>$= 120^\circ$<br>In $\triangle$ 's $PBA$ and $PAQ$<br>$\angle P$ is common<br>$\angle PAQ = 120^\circ$ (given)<br>$= \angle PBA$<br>$\therefore \triangle PBA \parallel \triangle PAQ$ (equiangular) | <b>3 marks</b><br>Correct solution including full reasoning.<br><b>2 marks</b><br>Substantial progress towards correct solution including full reasoning.<br><b>1 mark</b><br>Some progress towards correct solution. |
| <b>H2, H5</b> | <b>(ii)</b><br>Now, $\frac{PA}{PQ} = \frac{PB}{PA}$ (corresponding sides in similar triangles are in the same ratio)<br>$\therefore PA^2 = PQ \cdot PB$                                                                                                                                                                                                                                    | <b>1 marks</b><br>Correct solution.                                                                                                                                                                                   |

|                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                        |   |   |   |    |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Year 12 HSC                                                                                                                    | Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Trial Examination 2014 |   |   |   |    |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Question No. 16                                                                                                                | Solutions and Marking Guidelines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |   |   |   |    |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Outcomes Addressed in this Question                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                        |   |   |   |    |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                        |   |   |   |    |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Outcome                                                                                                                        | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marking Guidelines     |   |   |   |    |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| H5                                                                                                                             | <p>(a) Max. height <math>= 50 + \frac{40}{1 - 0.8} = 250 \text{ cm}</math></p> <p>(b) Common difference <math>= -5</math><br/>Therefore <math>T_1 = 42</math> and <math>T_{10} = -3</math><br/><math>S_{10} = \frac{10}{2}(42 - 3) = 195</math></p> <p>(c) (i) <math>A_1 = P(1.09) - 5000</math><br/>(ii) <math>A_3 = P(1.09)^3 - 5000(1 + 1.09 + 1.09^2)</math><br/>(iii)<br/><math>A_6 = 0 = P(1.09)^6 - 5000(1 + 1.09 + 1.09^2 + \dots + 1.09^5)</math><br/><math display="block">P = \frac{5000(1.09^6 - 1)}{0.09(1.09^6)}</math><br/><math>\\$P = \\$22429.59</math></p> <p>(d) (i)</p> <table><tr><td></td><td>1</td><td>3</td><td>5</td><td>7</td><td>9</td><td>11</td></tr><tr><td>2</td><td>Win</td><td>Lose</td><td>L</td><td>L</td><td>L</td><td>L</td></tr><tr><td>4</td><td>W</td><td>W</td><td>L</td><td>L</td><td>L</td><td>L</td></tr><tr><td>6</td><td>W</td><td>W</td><td>W</td><td>L</td><td>L</td><td>L</td></tr><tr><td>8</td><td>W</td><td>W</td><td>W</td><td>W</td><td>L</td><td>L</td></tr><tr><td>10</td><td>W</td><td>W</td><td>W</td><td>W</td><td>W</td><td>L</td></tr><tr><td>12</td><td>W</td><td>W</td><td>W</td><td>W</td><td>W</td><td>W</td></tr></table><br>$P(\text{Win}) = \frac{21}{36} = \frac{7}{12}$ <p>(ii) <math>P(\text{Exactly one win from two}) = \frac{7}{12} \times \frac{5}{12} + \frac{5}{12} \times \frac{7}{12}</math><br/><math display="block">= \frac{70}{144} \text{ or } \frac{35}{72}</math><br/>(iii) <math>P(\text{Win at least once}) = 1 - P(\text{lose both})</math><br/><math display="block">= 1 - \frac{5}{12} \times \frac{5}{12}</math><br/><math display="block">= \frac{119}{144}</math></p> |                        | 1 | 3 | 5 | 7  | 9 | 11 | 2 | Win | Lose | L | L | L | L | 4 | W | W | L | L | L | L | 6 | W | W | W | L | L | L | 8 | W | W | W | W | L | L | 10 | W | W | W | W | W | L | 12 | W | W | W | W | W | W | <p>(a) <b>1 mark</b> : correct answer.</p> <p>(b) <b>2 marks</b> : correct solution</p> <p><b>1 mark</b> relevant progress toward correct solution</p> <p>(c) (i) <b>1 mark</b> correct expression.<br/>(ii) <b>2 marks</b>: correct expression or equivalent.<br/><b>1 mark</b>: Significant progress.<br/>(iii) <b>2 marks</b>: Correct solution.<br/><b>1 mark</b>: Relevant progress, correct expression for <math>P</math>.</p> <p>(d) (i) <b>1 mark</b>: Using a method to show that the event occurs 21 ways out of 36</p> <p>(ii) <b>2 marks</b>: Correct solution.<br/><b>1 mark</b>: significant progress.<br/>(iii) <b>1 mark</b>: Correct answer..</p> |
|                                                                                                                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                      | 5 | 7 | 9 | 11 |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2                                                                                                                              | Win                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Lose                   | L | L | L | L  |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 4                                                                                                                              | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | W                      | L | L | L | L  |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 6                                                                                                                              | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | W                      | W | L | L | L  |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8                                                                                                                              | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | W                      | W | W | L | L  |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 10                                                                                                                             | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | W                      | W | W | W | L  |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12                                                                                                                             | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | W                      | W | W | W | W  |   |    |   |     |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |    |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

(e) (i)



(ii)  $P(\text{Only 1 passes}) = P(PF) + P(FP)$

$$= 0.75 \times 0.6 + 0.25 \times 0.4$$

$$= 0.55 \text{ or } 55\%$$

(e) (i) 2 marks: Weighted tree with branches labelled and outcomes listed.

1 mark: Tree with feature excluded.